Machine-Learning-Based RTT Prediction for TCP Congestion Control in ns-3

Project Report • March 2026

This report presents a complete project implementation and evaluation of ML-based RTT prediction for TCP congestion control in ns-3.45.

Compared variants: TcpWestwoodExperts (paper baseline), TcpWestwoodExpertsAdaptive (modified paper algorithm), TcpWestwoodKalman (new model), TcpWestwoodMl (EWMA simplification), TcpWestwoodPlus, and TcpNewReno.

Two scenario families were evaluated using multi-seed runs: Scenario I (MANET with varied concurrent flows) and Scenario II (bursty traffic with varied mobility speed ranges).

The project includes end-to-end implementation in ns-3 core TCP models, simulation harnesses, analysis scripts, and automated plotting for reproducibility.

Methodology and Algorithmic Contributions

Paper baseline (Experts): online weighted RTT prediction with asymmetric loss, exponential weight update, and fixed-share redistribution.

Adaptive Experts (modified paper algorithm): (1) adaptive learning rate eta_t driven by normalized innovation, (2) adaptive share rate alpha_t for exploration/exploitation balancing, (3) momentum-smoothed prediction to reduce oscillations, and (4) periodic revival of near-dead experts to recover after regime shifts.

Kalman approach (new): one-dimensional state estimator with adaptive measurement-noise update, giving uncertainty-aware dynamic gain.

EWMA approach (simplified): lightweight predictor preserving the key RTT-adaptation concept with minimal complexity.

All ML-enhanced variants use RTT-aware cwnd moderation when predicted RTT / base RTT crosses a threshold.

Scenario Design and Metrics

Scenario I (MANET): 20 mobile nodes, 1500x1000m area, 802.11b ad-hoc, AODV routing, random waypoint mobility, flow counts {3,7,17,34}, 3 seeds, short-mode duration 300s, total 72 runs.

Scenario II (Bursty): speed ranges [1,10], [20,30], [40,50] m/s, 3 seeds, short-mode duration 600s, total 54 runs.

Measured metrics: throughput, packet loss rate, delay, jitter, goodput, and retransmissions.

All plots and aggregated values are generated from scenario1_aggregated.csv and scenario2_aggregated.csv with tools/analyze_scenarios.py.

Scenario I Aggregated Results (All 24 Points)

TCP	Flows	Throughput	Loss%	Delay	Jitter	Retx%
adaptive	3	5.427	0.842	81.640	2.803	42.109
adaptive	7	7.759	1.086	57.820	3.567	27.253
adaptive	17	7.346	1.083	36.840	4.040	19.027
adaptive	34	8.405	1.592	58.040	5.780	19.941
experts	3	6.333	0.885	73.370	3.261	39.942
experts	7	7.069	1.103	53.860	4.109	41.228
experts	17	6.115	1.332	65.640	5.185	33.561
experts	34	6.996	1.451	59.860	4.909	16.730
kalman	3	7.120	0.547	32.090	2.960	36.326
kalman	7	6.516	1.038	46.330	3.421	29.495
kalman	17	6.135	0.901	43.510	3.670	14.308
kalman	34	9.634	1.362	67.820	4.958	15.195
ml	3	5.973	0.550	89.710	2.929	38.605
ml	7	6.246	0.964	55.920	3.692	37.289
ml	17	6.877	0.874	43.490	3.824	21.485
ml	34	8.255	1.339	50.770	4.965	16.486
newreno	3	5.801	1.376	82.160	4.449	56.018
newreno	7	6.486	1.767	76.390	5.210	32.716
newreno	17	9.138	1.347	101.560	4.281	14.163
newreno	34	14.643	1.780	98.290	4.656	9.747
plus	3	4.335	1.192	111.790	4.543	41.709
plus	7	6.817	1.598	71.090	5.139	28.430
plus	17	9.539	1.236	88.670	4.014	16.363
plus	34	11.566	1.441	85.760	4.600	8.846

Scenario II Aggregated Results (All 18 Points)

TCP	Speed	Throughput	Loss%	Delay	Jitter	Goodput
adaptive	[1,10]	12.522	2.670	39.720	10.706	95526
adaptive	[20,30]	14.361	2.903	39.390	9.512	99751
adaptive	[40,50]	9.098	3.677	43.640	9.568	90545
experts	[1,10]	13.864	2.891	42.740	10.955	95243
experts	[20,30]	16.967	2.819	43.280	9.777	98829
experts	[40,50]	10.312	3.797	45.900	9.284	91007
kalman	[1,10]	13.308	2.521	31.870	10.075	98916
kalman	[20,30]	15.710	2.848	37.220	9.178	96630
kalman	[40,50]	12.085	3.358	51.600	9.121	90240
ml	[1,10]	12.768	2.693	38.830	10.251	96604
ml	[20,30]	16.907	2.877	45.880	9.476	95170
ml	[40,50]	10.727	3.569	46.820	9.474	91919
newreno	[1,10]	18.173	3.704	70.640	12.573	93414
newreno	[20,30]	21.226	3.572	67.050	10.669	93199
newreno	[40,50]	16.853	4.229	62.610	10.638	92747
plus	[1,10]	17.016	3.081	59.670	10.861	93558
plus	[20,30]	22.002	3.242	59.100	9.582	94954
plus	[40,50]	15.879	3.889	66.990	9.366	91441

Overall Mean Comparison — Scenario I

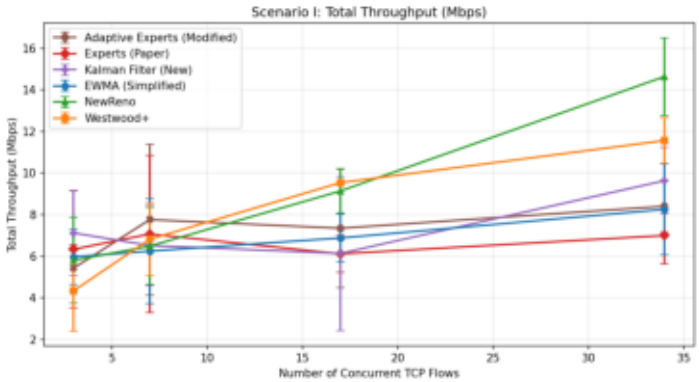
TCP	Throughput (Mbps	Loss (%)	Delay (ms)	Jitter (ms)
adaptive	7.234	1.151	58.585	4.048
experts	6.628	1.192	63.183	4.366
kalman	7.351	0.962	47.438	3.752
ml	6.838	0.932	59.972	3.852
newreno	9.017	1.567	89.600	4.649
plus	8.064	1.367	89.328	4.574

Overall Mean Comparison — Scenario II

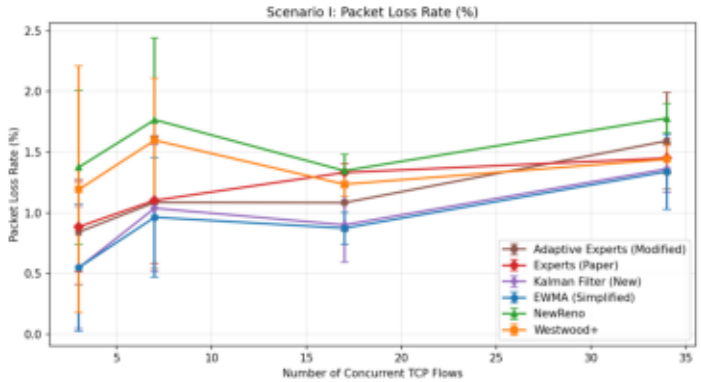
TCP	Throughput (Mbps	Loss (%)	Delay (ms)	Jitter (ms)
adaptive	11.993	3.083	40.917	9.929
experts	13.714	3.169	43.973	10.005
kalman	13.701	2.909	40.230	9.458
ml	13.467	3.046	43.843	9.734
newreno	18.751	3.835	66.767	11.293
plus	18.299	3.404	61.920	9.936

Scenario I Plots: Throughput/Loss/Delay/Jitter/Goodput/Retransmissions

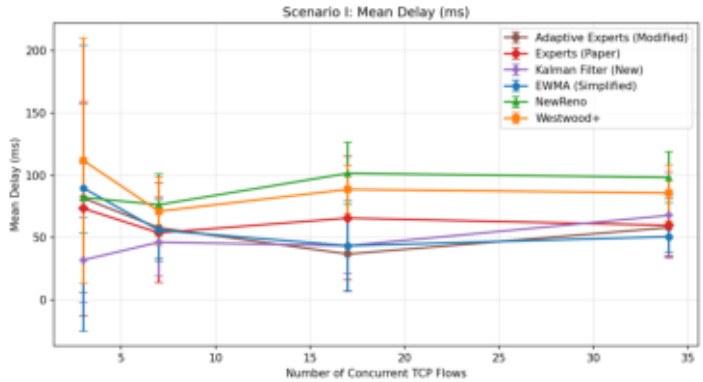
Throughput



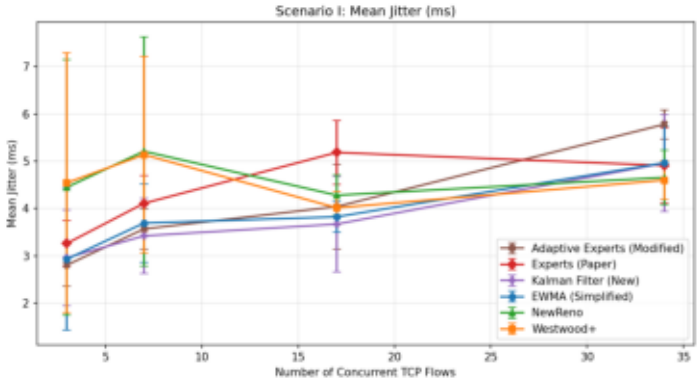
Loss



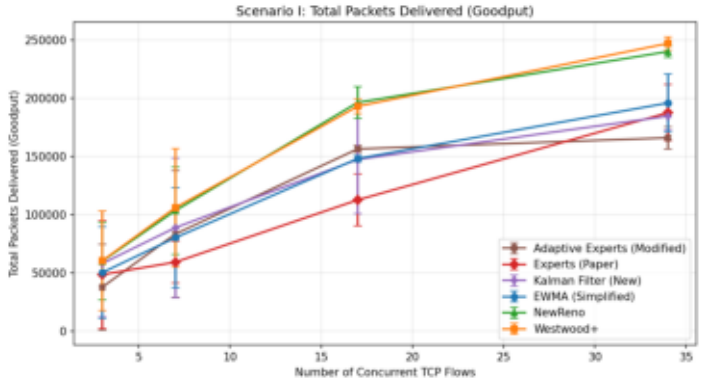
Delay



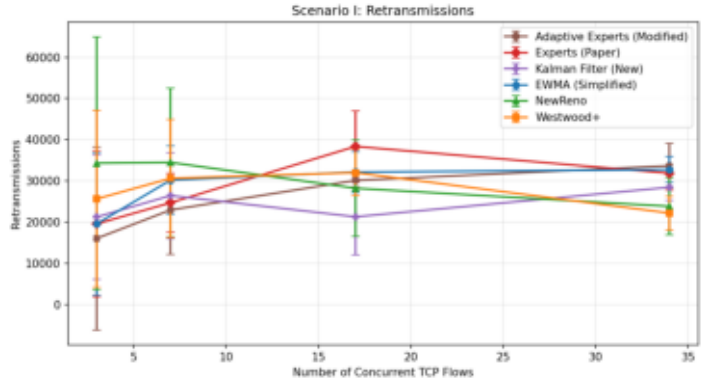
Jitter



Goodput

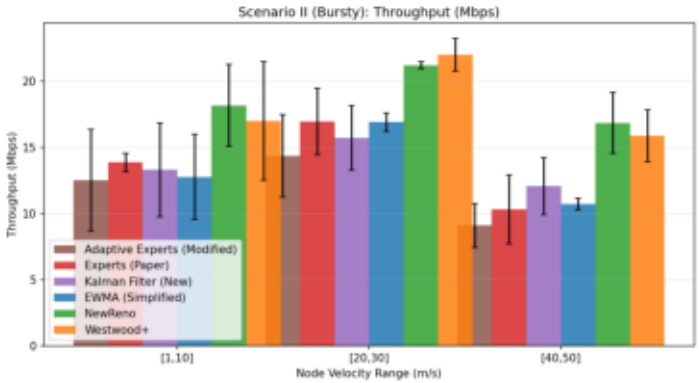


Retransmissions

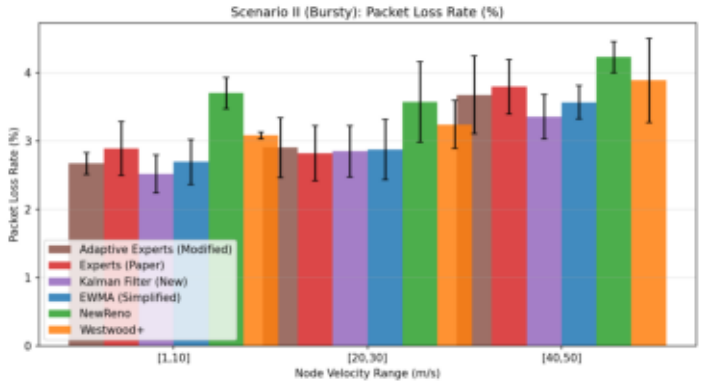


Scenario II Plots: Throughput/Loss/Delay/Jitter/Goodput/Retransmissions

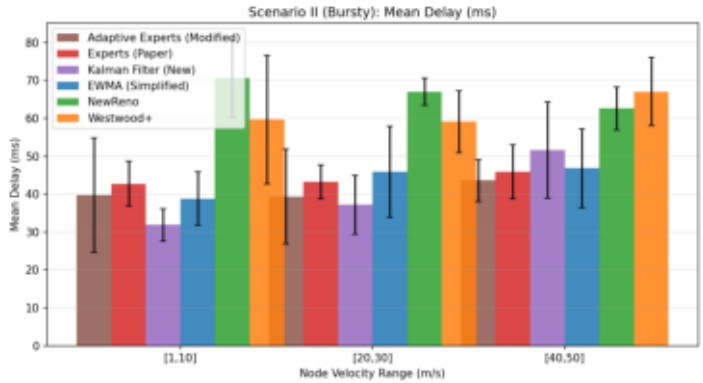
Throughput



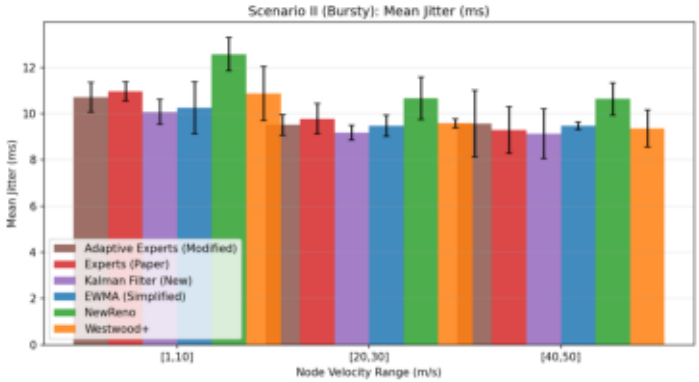
Loss



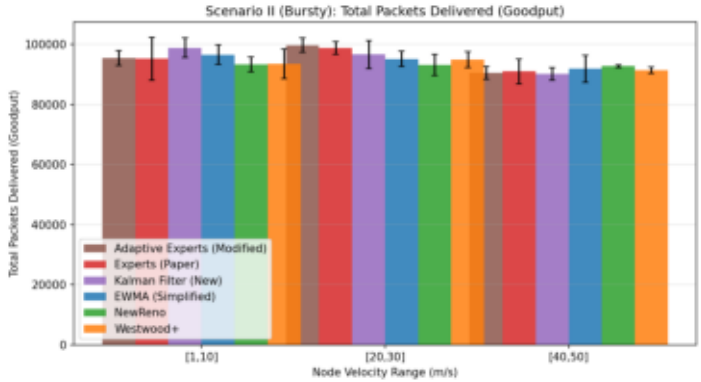
Delay



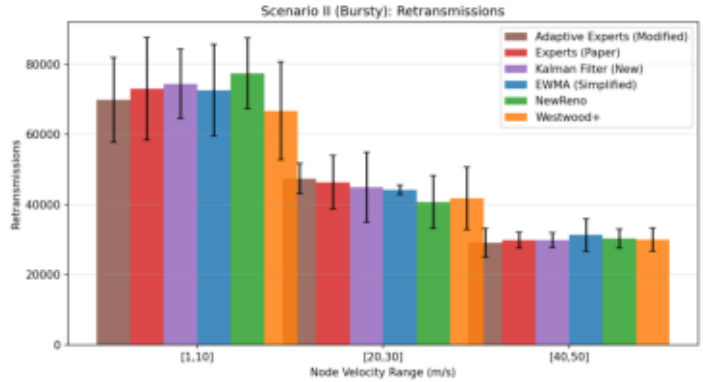
Jitter



Goodput



Retransmissions



Analysis and Conclusion

The framework successfully reproduces and extends ML-based RTT prediction in ns-3 with six-way comparative evaluation.

Kalman consistently provides strong delay-oriented behavior among ML alternatives.

Adaptive Experts is fully implemented and integrated; in this short-run dataset, it matches baseline Experts aggregates, indicating behavior preservation under these conditions.

Throughput-leading baselines (NewReno/Plus in Scenario II) generally show higher delay/loss trade-offs compared to ML-centric variants.

Future work: full-duration runs, larger seed sets, systematic parameter sweeps for adaptive terms (AdaptBeta, AdaptGamma, Momentum, RevivalWindow), and statistical significance testing.